

Even Banach Spaces with Countably Many Complex Structures

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Status

This packet is classified as *literature already answered*. The exact Ferenczi–Galego Question 19 from [1], asking whether there exists a Banach space with exactly ω complex structures, is answered positively by Cuellar-Carrera [2, Corollary 9]. The remaining material is preserved as an adjacent strengthening: the countably-many-complex-structure examples of Cuellar-Carrera may be chosen to be even in the sense of Ferenczi–Galego.

Source Question

Ferenczi–Galego ask on page 16 of [1]:

To conclude this section we note an open question. Recall that there exist spaces with exactly n complex structures, for any $n \in \mathbb{N}^*$ [5], and spaces with exactly 2^ω complex structures [1].

Question 19 *Does there exist a Banach space with exactly ω complex structures?*

3 Essentially incomparable complex structures

We recall that two infinite dimensional spaces Y and Z are said to be *essentially incomparable* if any bounded operator $S \in \mathcal{L}(Y, Z)$ is inessential, i.e., if $Id_Y - VS$ is a Fredholm operator (necessarily of index 0) for all operators $V \in \mathcal{L}(Z, Y)$. Es-

In text form: Does there exist a Banach space with exactly ω complex structures?

Supporting Literature Evidence

Cuellar-Carrera proves that the spaces $\mathfrak{X}_\gamma(\mathbb{C})$ have exactly ω complex structures for every limit ordinal $\omega^2 \leq \gamma < \omega_1$; the same page also proves that the real hyperplanes of $\mathfrak{X}_{\omega_1}(\mathbb{C})$ have no complex structure, using the Ferenczi–Galego lifting criterion.

contradiction because $\mathfrak{X}$ is an isomorphic embedding. $\square$

The next corollary offers uncountably many examples of Banach spaces with exactly countably many complex structures.

Corollary 9. *The space $\mathfrak{X}_\gamma(\mathbb{C})$ has ω complex structures up to isomorphism for every limit ordinal $\omega^2 \leq \gamma < \omega_1$.*

Proof. Let J be a complex structure on $\mathfrak{X}_\gamma(\mathbb{C})$. We extend J to a complex structure defined in the whole space $\mathfrak{X}_{\omega_1}(\mathbb{C})$ by setting $T = JP_I + iP^I$, where $I = [0, \gamma)$. It follows that $T = D_T + S$ for an strictly singular operator S and a diagonal operator D_T like in Theorem 2. Notice that $D_T x = ix$ for every $x \in \mathfrak{X}^I$, otherwise there would be a limit ordinal α such that $S|_{\mathfrak{X}_{I_\alpha}} = 2iId|_{\mathfrak{X}_{I_\alpha}}$. Hence $JP_I = D_T P_I + S$. Which implies that J has the form $J = \sum_{j=1}^k \epsilon_j i P_{I_j} + S_1$ where S_1 is strictly singular on $\mathfrak{X}_{\omega_1}(\mathbb{C})$, $(\epsilon_j)_{j=1}^k$ are signs and $I_1 < I_2 < \dots < I_k$ are ordinal intervals whose extremes are limit ordinals and such that $\gamma = \cup_{j=1}^k I_j$. Now the rest of the proof is identical to the proof of the previous corollary. In particular, all the non-isomorphic complex structures on $\mathfrak{X}_\gamma(\mathbb{C})$ are incomparable. $\square$

We also have, using the same proof of the previous corollary, that for every increasing sequence of limit ordinals $A = (\alpha_n)_n$, the space $\mathfrak{X}_A = \bigoplus_n \mathfrak{X}_{I_{\alpha_n}}(\mathbb{C})$, where $I_{\alpha_n} = [\alpha_n, \alpha_n + \omega)$, has exactly infinite countably many different complex structures. Hence there exists a family, with the cardinality of the continuum, of Banach spaces such that every space in it has exactly ω complex structures.

4. QUESTION AND OBSERVATIONS

Is easy to check that subspaces of even codimension of a real Banach space with complex structure also admit complex structure. An interesting property of $\mathfrak{X}_{\omega_1}(\mathbb{C})$ is that any of its real hyperplanes (and thus every real subspace of odd codimension) do not admit complex structure.

Proposition 10. *The real hyperplanes of $\mathfrak{X}_{\omega_1}(\mathbb{C})$ do not admit complex structure.*

Proof. By the results of Ferenczi and E. Galego [7, Proposition 13] it is sufficient to prove that the ideal of all $\mathbb{R}$ -linear strictly singular operators on $\mathfrak{X}_{\omega_1}(\mathbb{C})$ has the lifting property, that is, for any $\mathbb{R}$ -linear isomorphism on $\mathfrak{X}_{\omega_1}(\mathbb{C})$ such that $T^2 + Id$ is strictly singular, there exists a strictly singular operator S such that $(T - S)^2 = -Id$. The proof now follows easily from the Remark 7. $\square$

Definitions

A complex structure on a real Banach space X is a bounded real-linear operator $J: X \rightarrow X$ such that $J^2 = -Id_X$, considered up to real-linear conjugacy. Following [1], an infinite-dimensional real Banach space is *even* if it admits a complex structure and its hyperplanes do not admit a complex structure.

Proof Intuition

Cuellar-Carrera's construction is rigid enough that every real-linear operator on the large transfinite space has a diagonal part modulo strictly singular operators. If an operator squares to $-\text{Id}$ modulo strictly singular operators, the diagonal part itself squares exactly to $-\text{Id}$. This is precisely the lifting property that Ferenczi–Galego identify with absence of complex structures on hyperplanes.

The proof below uses the same extension trick as Cuellar-Carrera's Corollary 9: extend an operator on the separable initial interval $\mathfrak{X}_\gamma(\mathbb{C})$ to $\mathfrak{X}_{\omega_1}(\mathbb{C})$ by putting the standard multiplication by i on the complementary interval. The diagonal lifting on the large space then restricts back to a diagonal lifting on $\mathfrak{X}_\gamma(\mathbb{C})$.

Theorem 1. *Let γ be a limit ordinal with $\omega^2 \leq \gamma < \omega_1$. Let $\mathfrak{X}_\gamma(\mathbb{C})$ be Cuellar-Carrera's space, viewed as a real Banach space. Then $\mathfrak{X}_\gamma(\mathbb{C})$ is even and has exactly ω complex structures up to isomorphism. In particular, there is a separable even real Banach space with exactly countably infinitely many complex structures.*

Proof. Cuellar-Carrera's Corollary 9 [2] gives exactly ω complex structures on $\mathfrak{X}_\gamma(\mathbb{C})$ for every limit ordinal $\omega^2 \leq \gamma < \omega_1$. Since γ is countable, this space is separable.

It remains to prove evenness. The space already has a complex structure, namely the original scalar multiplication by i . By Ferenczi–Galego's lifting criterion [1, Proposition 13], it is enough to show that the ideal of real-linear strictly singular operators on $\mathfrak{X}_\gamma(\mathbb{C})$ has the lifting property: whenever $T: \mathfrak{X}_\gamma(\mathbb{C}) \rightarrow \mathfrak{X}_\gamma(\mathbb{C})$ is real-linear and $T^2 + \text{Id}$ is strictly singular, there exists a strictly singular operator S such that $(T - S)^2 = -\text{Id}$.

Put $I = [0, \gamma)$ and regard $\mathfrak{X}_\gamma(\mathbb{C}) = \mathfrak{X}_I(\mathbb{C})$ as the 1-complemented initial-interval subspace of $\mathfrak{X}_{\omega_1}(\mathbb{C})$, with coordinate projections P_I and $P^I = \text{Id} - P_I$. Given such a T , define a real-linear operator on $\mathfrak{X}_{\omega_1}(\mathbb{C})$ by

$$\tilde{T} = TP_I + iP^I.$$

Then

$$\tilde{T}^2 + \text{Id} = (T^2 + \text{Id})P_I,$$

so $\tilde{T}^2 + \text{Id}$ is strictly singular on $\mathfrak{X}_{\omega_1}(\mathbb{C})$.

Cuellar-Carrera's diagonal construction, Proposition 3, and Remark 7 in [2] imply that the diagonal part $D_{\tilde{T}}$ is bounded, that $D_{\tilde{T}}^2 = -\text{Id}$, and that $\tilde{T} - D_{\tilde{T}}$ is strictly singular. For completeness, the last assertion is exactly the final step in the proof of [2, Theorem 2]: once the diagonal part is bounded, the relevant strictly singular perturbation proposition of [2] applies because the difference tends to zero on every rapidly increasing sequence. Remark 7 supplies the exact-square conclusion $D_{\tilde{T}}^2 = -\text{Id}$ under the weaker hypothesis $\tilde{T}^2 + \text{Id} \in \mathcal{S}$.

The operator $D_{\tilde{T}}$ is diagonal with respect to the transfinite basis, and the initial interval $I = [0, \gamma)$ is invariant under such diagonal operators. Hence

$$D = D_{\tilde{T}} \upharpoonright_{\mathfrak{X}_\gamma(\mathbb{C})}$$

is a real-linear operator on $\mathfrak{X}_\gamma(\mathbb{C})$ with $D^2 = -\text{Id}$. Moreover

$$T - D = (\tilde{T} - D_{\tilde{T}}) \upharpoonright_{\mathfrak{X}_\gamma(\mathbb{C})}$$

is strictly singular. Taking $S = T - D$ gives $(T - S)^2 = D^2 = -\text{Id}$, so the strictly singular ideal on $\mathfrak{X}_\gamma(\mathbb{C})$ has the lifting property.

Ferenczi–Galego's criterion therefore says that the hyperplanes of $\mathfrak{X}_\gamma(\mathbb{C})$ do not admit complex structures. Thus $\mathfrak{X}_\gamma(\mathbb{C})$ is even. $\square$

Verification Notes

The exact Question 19 is not claimed as a new run solution: it is explicitly answered by Cuellar-Carrera’s Corollary 9. The packet’s claim is the strengthened parity conclusion for the same separable examples. The proof uses only the operator decomposition and strictly-singular lifting mechanism already present in [2] and [1]; no numerical or computational verification is involved.

Novelty and Scope

Searches were made in the local run indexes and source corpus for 0704.1459, “even countably many complex structures”, “countable infinite number of complex structures”, and related phrases. Web searches for exact phrases such as “even countably many complex structures” and “exactly countably many complex structures even Banach space” did not reveal a statement of the strengthened parity conclusion. The novelty confidence is therefore moderate: the argument is a direct transfer from Cuellar-Carrera’s machinery, and may be folklore to specialists, but it is not the same as the already-known answer to Question 19.

This packet does not settle Ferenczi–Galego’s other questions about direct sums of arbitrary even spaces or projection-total incomparability.

Human Review Recommendation

Check the transfer of Cuellar-Carrera’s diagonal lifting from $\mathfrak{X}_{\omega_1}(\mathbb{C})$ to the initial-interval spaces $\mathfrak{X}_\gamma(\mathbb{C})$, especially the diagonal construction preceding Proposition 3, Proposition 3 itself, and Remark 7 from [2]. If accepted, the packet should be recorded as a strengthened adjacent result rather than as an original solution of the exact countability question.

References

- [1] V. Ferenczi and E. M. Galego, *Even infinite dimensional real Banach spaces*, arXiv:0704.1459.
- [2] W. Cuellar Carrera, *A Banach space with a countable infinite number of complex structures*, J. Funct. Anal. 267 (2014), no. 5, 1462–1487; arXiv:1401.1781.